Technical Report: Indian Bird Species Identification

1. Dataset & Preprocessing

The dataset consists of thousands of images across 25 categories. To ensure the model learned general features rather than memorizing the sequence of the data, the following steps were implemented:

Resolution: All images were resized to **224x224** to maintain compatibility with standard pre-trained architectures.

Data Shuffling: A critical step where the data was randomized before splitting into Training and Validation sets. Initial tests without shuffling resulted in near-zero validation accuracy.

Augmentation: Rescaling was applied to normalize pixel values between 0 and 1, facilitating faster gradient descent convergence.

2. Model Architectures

Two distinct approaches were developed to find the optimal balance between accuracy and computational efficiency:

a) **Custom CNN (From Scratch)**
- **Architecture:** 4 Convolutional blocks followed by MaxPooling, Batch Normalization, and Dropout (0.5).
- **Design Intent:** To build a specialized feature extractor specifically for bird anatomy.
- **Outcome:** Highly prone to overfitting. While the training accuracy reached **95.14%**, the validation accuracy remained stagnant at **60.65%**, indicating the model was "memorizing" specific training images rather than learning bird species traits.

b) **Transfer Learning (MobileNetV2)**
- **Architecture:** Pre-trained MobileNetV2 backbone (ImageNet weights) with a custom Global Average Pooling head and a 512-unit Dense layer.
- **Design Intent:** To utilize "frozen" weights from a model already trained on 1.4 million images to extract high-level features like textures, edges, and shapes.
- **Outcome:** Superior generalization. Achieved **98.11%** Training Accuracy and **91.54%** Validation Accuracy.

3. Performance Comparison

Metric	**Custom CNN**	**Transfer Learning (MobileNetV2)**
Training Accuracy	95.14%	**98.11%**
Validation Accuracy	60.65%	**91.54%**
Validation Loss	1.98	**0.27**
Generalization Gap	~35% (High Overfit)	**~6% (Robust)**

4. Evaluation & Error Analysis

The MobileNetV2 model showed exceptional performance on distinctive species like the **Cattle Egret (0.99 F1-score)**. Some confusion was noted in species with similar plumage patterns (e.g., Wagtail variants), suggesting that while the model is highly accurate, fine-grained classification remains the most challenging aspect of bird identification.

5. Conclusion

The experimental results confirm that **Transfer Learning** is the most effective strategy for multi-class avian classification. By utilizing pre-trained feature extractors, the system achieved a **30.89% increase in validation accuracy** compared to the custom-built model. The final model is lightweight, accurate, and ready for deployment in real-time identification environments.